

CDS 3543 – Data Visualization for Decision Making

Week 14: Lab 12
Case Study - Global Terrorism

Course Learning Outcome: CLO4

Chapters:

Chapter 8: Case Studies

Cognitive Levels: Apply

Aim: In this lab, you will analyze the Global Terrorism Dataset to uncover trends and insights related to terrorism incidents worldwide. You will engage in data cleaning, preparation, analysis, and visualization using Python.

Objectives

- Understand the significance of analyzing terrorism data.
- Learn techniques for data cleaning and preparation.
- Explore methods for visualizing trends in the data.
- Interpret findings to derive meaningful insights.

Tools Required: Python

Requirements:

- Basic knowledge of Python and pandas.
- Familiarity with data visualization libraries (e.g., Matplotlib, Seaborn).

Document Revision Control:

Version	Author	Effective Date	Change Description	DRC No
1.0	Habeeb Shah Khan	202510	Defining the first version	00X

Lab Steps

Step 1: Data Gathering

1. Download the Dataset:
 - Download the Global Terrorism Database. ([global_terrorism.csv](#))
 - Extract the dataset files to your working directory.

Step 2: Data Loading

1. Import Necessary Libraries:
2. Load the Dataset:
3. Display Basic Information:

Step 3: Data Cleaning

1. Identify Missing Values:
2. Remove Rows with Critical Missing Values:
3. Confirm Data Cleaning:

Step 4: Analyzing Terrorism Trends

1. Identify Unique Attack Types:
2. Count Incidents by Attack Type:

Step 5: Visualizing Yearly Trends

1. Calculate Total Incidents Over Years:
2. Plot Yearly Trends:

Step 6: Countries with Highest Incidents

1. Analyze Incidents by Country:
2. Visualize Top 10 Countries:

Step 7: Most Common Attack Types

1. Visualize Leading Attack Types:

Step 8: Interpretation of Findings

Summarize your findings:

1. What are the total incidents recorded?
2. What significant trends did you observe?
3. Discuss implications for national security policies.

Step 9: Conclusion

1. Recap the analysis process and gained insights.
2. Reflect on the importance of data-driven decision-making in understanding terrorism.

Submission

- Submit your Jupyter Notebook or script with all code and findings documented.